Color-Invariant Saree Pattern Recognition

Portfolio Project Summary

Objective

Develop a computer-vision pipeline that represents saree images as compact feature embeddings and retrieves visually similar patterns using cosine similarity. Colour augmentation is used to reduce reliance on colour and encourage attention to pattern structure.

Task Objective

Given an RGB saree image, rank reference images by similarity and support pairwise verification using embedding similarity. The intended challenge is to recognise a design despite colour variations while distinguishing different motifs. The available dataset labels support pattern-category evaluation; they do not establish same-design identity across colourways.

Dataset

Indian Saree Patterns dataset from Kaggle (div456/indian-saree-patterns). The notebook detected 1,468 readable images across the supplied train/validation/test folders: 1,293 training, 115 validation and 60 test images. Four pattern categories were identified: Banarasi, Bandhani, Ikat and Pichwai. Dataset source: https://www.kaggle.com/datasets/div456/indian-saree-patterns

Approach & Deliverables

A pretrained EfficientNet-B0 backbone feeds a 256-dimensional L2-normalised embedding head and a four-class classifier. Training uses label-smoothed cross-entropy with AdamW, learning-rate scheduling, early stopping, and colour-focused augmentations including colour jitter, hue/saturation shifts and grayscale. Cosine similarity is used for gallery ranking and pair verification.

Notebook deliverables include model training and inference, classification evaluation, category-level gallery/query retrieval, pair verification metrics, sample retrieval visualisations, and a basic CPU efficiency measurement.

Outcome (from the executed notebook)

- Test classification accuracy: 91.67% (60 test images); best validation accuracy: 95.65%.
- Category-level retrieval: Recall@1 = 85.00%, Recall@3 = 95.00%, Recall@5 = 97.50%, mAP = 89.49%.
- Category-level pair verification: ROC-AUC = 0.9502; EER = 0.1050.
- Model: EfficientNet-B0; 4.40 million trainable parameters; 256-D embeddings; measured single-image CPU latency ≈ 43.19 ms.

Scope & Limitation

These results measure four-category pattern recognition/retrieval, not exact product-level or same-motif matching across colourways. A stronger colour-invariance claim requires identity-level labels or paired examples of the same design in different colours, plus a split and evaluation protocol designed around those identities. The notebook and summary are independent portfolio materials; no company name, internal instructions, or confidential source document is included.